Capillary Electrophoresis Products

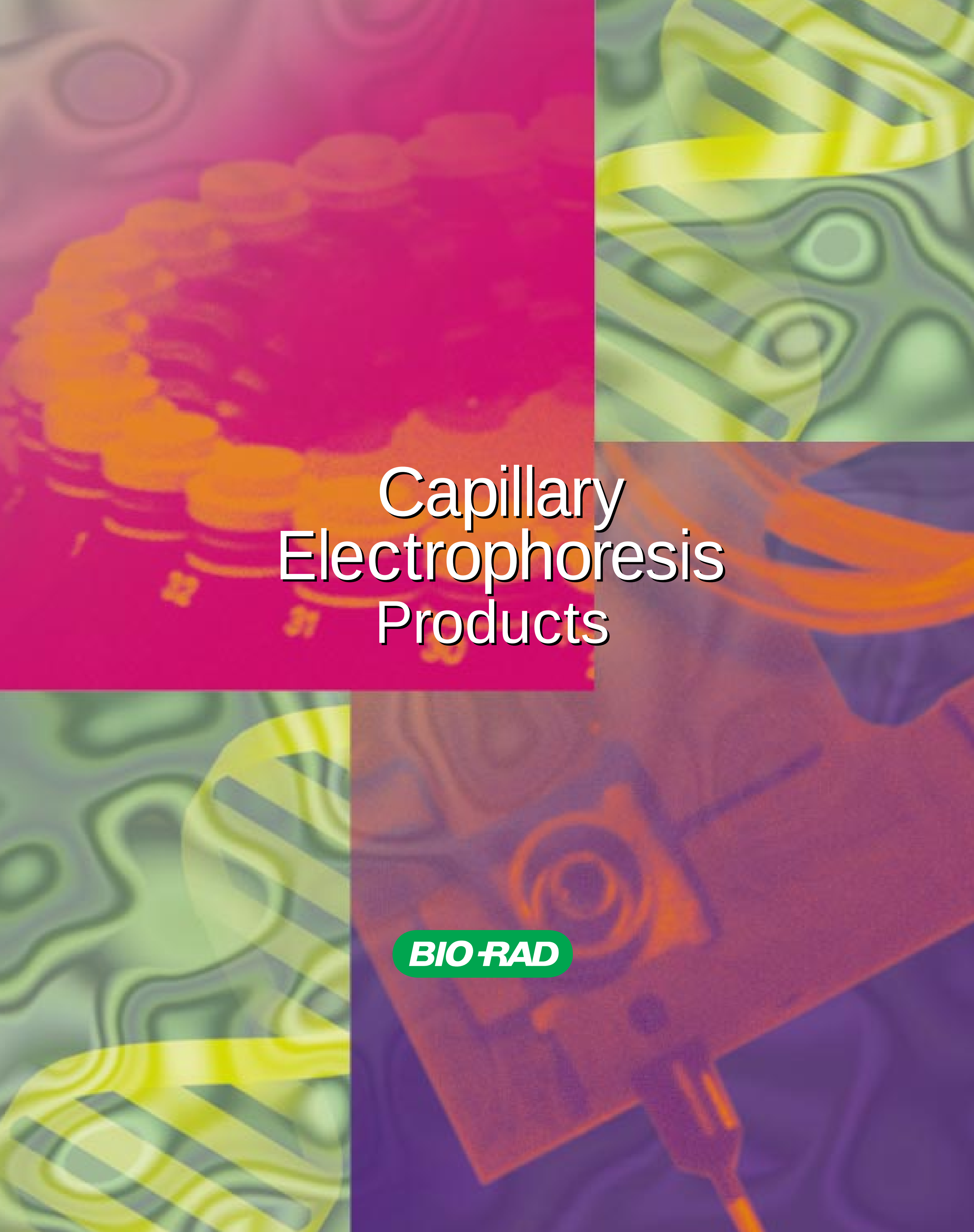


Table of Contents

New Products 2

BioFocus Automated Systems 4

Laser Induced Fluorescence Detection 8

Fluorescence Kits 10

CE/Mass Spectrometry 11

Capillaries and Cartridges 12

Kits and Reagents 14

Application Notes Index 20

New BioFocus LIF2

Dual Laser Fluorescence Detection System

- Remote laser module with fiber optic connections
- Accommodates two lasers simultaneously for dual wavelength excitation, 488 nm argon-ion and 594 nm helium neon
- Dual PMT configuration for tandem detection of two separate emission wavelengths
- Increased detection sensitivity over conventional UV absorbance down to the picomole level for DNA
- Accepts one external laser source in place of the helium neon laser
- Three different configurations to match your needs and budget

New dsDNA 1000

Fluorescent DNA Analysis Kit

- Fluorescent analysis of dsDNA from 100 to 1,000 base pairs
- High resolution dynamic sieving separations
- Easy and rapid labeling
- Superior fluorescence sensitivity
- Highly reproducible migration times and peak areas

New BioFocus Verification Kit

Automated Performance Verification

- Simple assessment of system performance
- Rapid and accurate diagnosis of all important parameters
- Automated reporting capability
- Comprehensive kit with capillaries and all required reagents

New CE Oligonucleotide Kit

Purity Analysis

- Fast and efficient purity analysis of synthetic oligonucleotides
- Simple automated protocol
- Reproducible high resolution results
- Unique polymer-coated BioCAP capillary eliminates EOF
- Single base resolution of oligonucleotides from 4 to 40 mer

New BioMark

Synthetic pI Markers

- Accurate pI standards for CIEF
- Highly stable, producing sharp peaks
- Water soluble and ideal for use as internal standards
- Strong absorbance in both the UV and visible regions
- Range of 16 pI values from 5.3 to 10.4
- Available as kits or individually

Capillary Electrophoresis

Capillary electrophoresis allows you to automate the analysis of a wide range of compounds, including peptides, proteins, nucleic acids, oligonucleotides, and pharmaceuticals. This section describes the family of capillary electrophoresis products, which includes capillary electrophoresis instruments, capillaries, reagents, and data handling devices.

BioFocus Automated Systems

Bio-Rad offers two fully automated capillary electrophoresis systems. The high performance, high value BioFocus 2000 system is ideal for routine analysis. It provides complete automation, with capillary temperature control and high-sensitivity programmable single wavelength UV detection, and offers optional visible detection, and autosampler temperature control. The full-featured BioFocus 3000 high performance system for fast, automated methods development and analysis, provides complete automation with capillary temperature control and high-sensitivity single wavelength, multi-wavelength, and spectral scanning UV/Vis detection. Autosampler temperature control is optional.

High Sensitivity

Both BioFocus CE systems are equipped with high-sensitivity, low-noise UV/Vis detectors and provide quiet and stable baselines.

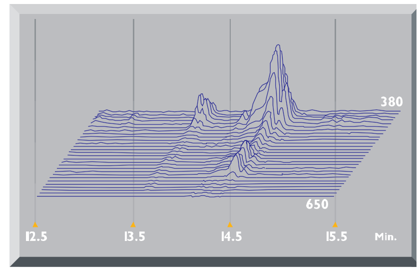
BioFocus 3000 DETECTOR SENSITIVITY DATA

	Benzoate @ 200 nm	Tryptophan @ 280 nm
Baseline noise	25 µAU	25 µAU
Sensitivity	30 AU/M	42 AU/M
Detection limit	3 µm	2 µm

Note: Typical performance values using a 50 µm ID capillary.

Fast UV/Vis Scanning

The fast-scanning UV/Vis detector of the BioFocus 3000 system can monitor multiple wavelengths simultaneously. Three dimensional plots of absorption spectra versus migration time allow confirmation of peak purity and aid in peak identification.



Analysis of human hemoglobin by CIEF with fast-scanning visible detection provides additional information about sample components.

Excellent Reproducibility

Both BioFocus systems provide complete automation, consistent capillary temperature control, and precise pressure and electrokinetic injection. This results in highly reproducible separations with relative standard deviations (%RSDs) of typically <0.6% for migration times and 1–2% for peak areas.

Versatility

Removable 32-position inlet and outlet carousels allow greater flexibility in experimental design for multiple inlet samples and reagents and multiple outlet fractions, buffers, and wastes. Random access to all carousel vial positions permits flexible methods development. The BioFocus capillary cartridge allows easy interchange of coated, uncoated, gel-filled, or other capillaries of any length between 24–100 cm and 360–375 µm OD.

High Pressure Capillary Purge

The 100 psi (7 bar) purge capability permits rapid capillary conditioning and cleansing between runs, and enables the use of high viscosity buffers for the best resolution in dynamic sieving applications.

Capillary Temperature Control

Accurate capillary temperature control is critical for achieving reproducible results and is a standard feature on all BioFocus systems. The BioFocus capillary cartridge has a unique design that circulates liquid coolant at a high velocity in direct contact with the capillary, enabling faster and more efficient heat dissipation than air-cooled systems. Peltier thermo-electric units contained within the instrument are used to control the temperature of the capillary coolant.

Autosampler Temperature Control

Optional heating or cooling of the autosampler compartment is achieved with Peltier thermo-electric units independent of the capillary temperature control system. This system is completely enclosed within the BioFocus instrument and requires no external water chillers.

Automation

The BioFocus systems allow linking of multiple methods, automatic integration, and report printing for extended unattended operation. Programmed run sequences can be interrupted to run a priority sample, and then returned to the scheduled run sequence.

Software

Each BioFocus system includes operating and data analysis software. All software runs in Windows 3.1 or Windows 95. The graphical user interface and turnkey method templates make learning quick and easy. In addition to the detector data, the unalterable data files contain the run history of the voltage, current, and temperature, as well as instrument parameters and operator information. Data can be ported to any information system conforming to AIA standards.

Automatic Performance Verification

Every BioFocus system comes with a performance verification kit and software that provides automated verification test reports. This tool is used to verify system performance at installation, during periodic checks to confirm optimal system performance, and to facilitate troubleshooting. Kit components have been carefully designed to separate instrument performance from chemistry and capillary column considerations.

COMPUTER AND SOFTWARE REQUIREMENTS

Hardware Requirements *

Type	IBM PC (or 100% compatible) Pentium class
Total RAM	16 MB minimum
Disk drives	500 MB or greater recommended, 1.44 megabyte, 3.5", internal floppy disk drive required
Video system	VGA (or 100% compatible) color graphics or higher
Serial ports	Two RS-232 ports for asynchronous communications, configured as COM 1 (IRQ4, I/O Port Address 03F8) and COM 2 (IRQ3, I/O Port Address 02F8)
Mouse	Microsoft (or 100% compatible) BUS mouse (IRQ5) (must not use IRQ 0,1,2,3,4,7 or LPT1 IRQ)

**All hardware documentation, including computer and adaptor card settings, must be available.*

Hardware Options

Parallel port	One Centronix (or 100% compatible) for a printer (IRQ7) configured as LPT1
Printer	Centronix (or 100% compatible) parallel interface. Printer memory should be capable of processing full page graphics at 300 dpi.

Operating System Requirements

MS (or PC) DOS	Version 3.3x, Version 5.x, Version 6.0, 6.1, 6.2
Environment requirements	Microsoft Windows 3.1, 3.11, Windows for Workgroups 3.11, or Windows 95

SPECIFICATIONS

BioFocus Instrument

PC operation	IBM-compatible Pentium class computer with Windows 3.1x or Windows 95 software
Automated sampling and fraction collection	Two 32-position removable carousels for samples, buffers, washes, fractions, and wastes (can hold up to 31 samples and collect up to 30 fractions). Uses standard 500 µl microcentrifuge tubes, or custom vials with evaporation control caps; 500 µl, 1.5 ml, 4 ml
Detector	
BioFocus 2000 system	Programmable, single wavelength UV (190–365 nm) detector with optional programmable, single wavelength Vis (366–800 nm). Purchase of tungsten lamp required for visible range
BioFocus 3000 system	Programmable, fast-scanning UV/Vis (190–800 nm) detector allows single wavelength, multiple wavelength (up to 32 wavelengths), and high speed scanning detection
Power supply	Up to 30 kV constant voltage or 300 µA constant current; current and voltage history recorded. Polarity automatically reversible in software
Injection modes	Pressure, electrokinetic
Capillary cartridge	Accepts 360–375 µm OD capillary columns from 24 to 100 cm long. Provides liquid cooling for capillary. Fast, easy assembly. Has pre-aligned optics and requires no glues or adhesives
Temperature control	Programmable for capillary cartridge (15–40 °C) and autosampler compartment (4–40 °C) Autosampler temperature control is optional and field upgradable
Dimensions	19.5" wide x 23" deep x 22" high (49 cm x 58 cm x 56 cm)
Weight	
BioFocus 2000 system	104 lb (47 kg) without sample temperature control 130 lb (59 kg) with sample temperature control
BioFocus 3000 system	108 lb (49 kg) without sample temperature control 134 lb (61 kg) with sample temperature control
Power requirements	100, 120, 220, or 240 VAC, 50 or 60 Hz (900 watts minimum for BioFocus 3000 or BioFocus 2000 system with full temperature control; 500 watts minimum for BioFocus 2000 system without autosampler temperature control)
Ambient temperature	10–40 °C
Analog output	10, 100, or 1,000 mV for strip chart recorder, integrator, or PC workstation
Other requirements	Inert compressed gas required for capillary purging and pressure loading, 100 psi

Ordering Information

CATALOG NO.	PRODUCT DESCRIPTION
BioFocus 2000 Capillary Electrophoresis Systems	
148-0206	With sample temperature control, without computer
148-0236	With sample temperature control and computer
148-0207	Without sample temperature control, without computer
148-0237	Without sample temperature control, with computer
BioFocus 3000 Capillary Electrophoresis Systems	
148-0204	With sample temperature control, without computer
148-0234	With sample temperature control and computer
148-0205	Without sample temperature control, without computer
148-0235	Without sample temperature control, with computer
BioFocus Replacement Parts	
148-6050	Fluorinert FC 77 High Voltage Coolant, 118 ml
148-6059	Coolant Reservoir Filter, 60 µm, 5
167-0905	Deuterium Lamp
167-0906	Tungsten Lamp
148-6057	Pressure Chamber with Inlet Electrode
148-6058	Outlet Electrode
148-6060	Pneumatic Tubing and Fittings
148-6062	Vial Extraction Tool
148-6063	Outlet Electrode Changing Tool
148-6065	Capillary Purge Device
148-6066	Coolant Line Purge Device
148-6067	Coolant Reservoir Lid
148-6069	BioFocus System Instruction Manual
148-6071	Test Cartridge
148-6085	BioFocus Inlet Electrode Pack
New 148-6086	Inlet Carousel
New 148-6087	Outlet Carousel
New 148-6088	500 µl Vial Holder, 5
New 148-6089	2.0 ml Vial Holder, 5
New 148-6090	4.0 Vial Holder, 5
New 148-6091	Evaporation Control Vials, 500 µl, 500
New 148-6092	Evaporation Control Caps, 500

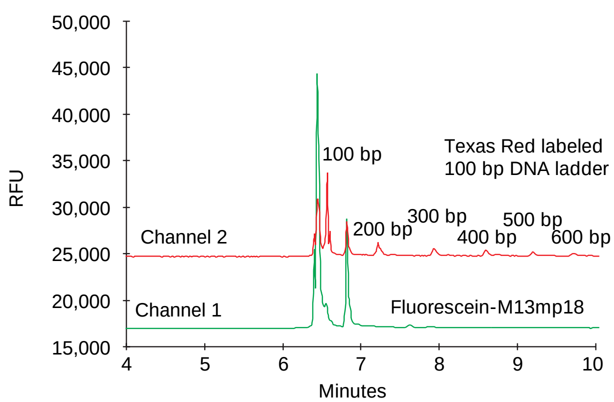
BioFocus® LIF² Capillary Electrophoresis System with Dual Laser Detection

The Only System of its Kind

Introducing the BioFocus LIF² detector, the only capillary electrophoresis system available with dual laser detection. This innovative design delivers two separate wavelengths of high intensity laser light to the sample simultaneously. The advantage is obvious—you get significant increases in detection sensitivity over conventional UV absorbance while detecting up to two separate fluorescent emission wavelengths simultaneously.

DNA Detection with Capillary Electrophoresis

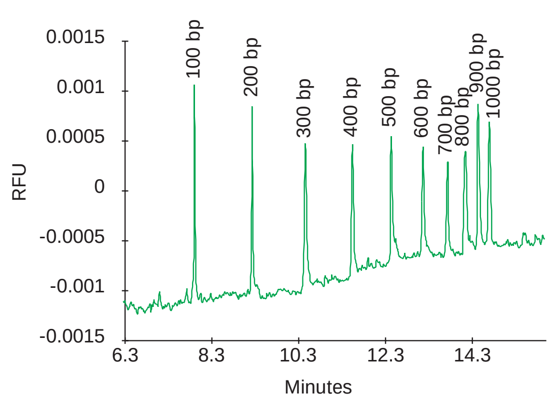
If you're performing PCR amplification, restriction fragment analysis, microsatellite analysis, or mutation analysis (SSCP), the BioFocus LIF² system can streamline your sample analysis. It's a fully-automated, high-sensitivity analysis system that performs high resolution separations of DNA fragments. In combination with the BioFocus dsDNA fluorescent detection kit, the BioFocus LIF² detector automatically analyzes multiple DNA samples with the highest resolution and sensitivity.



Two channel detection with internal standards.

The LIF² Detector: Two Lasers are Better than One

The heart of the LIF² detection system is the 488 nm argon-ion and 594 nm helium neon lasers which deliver laser light to the capillary via fiber optics. The multiple fluorescent signals are detected by photo multiplier tubes and the analog signal is digitized by the laser module. The LIF² detector is available in three different configurations—one laser and one data collection channel, one laser and two data collection channels, or two lasers and two data collection channels. The BioFocus LIF² system can even accept an external laser light source in place of the helium neon laser.



High sensitivity 4.2 zeptomole (x 10^{-21}) injected per band.

Superior Fluorescence Sensitivity

The BioFocus LIF² detector is at least 1,000 times more sensitive than existing UV detection. The key to laser induced fluorescence sensitivity is focusing a beam of high intensity, laser light on a capillary. The BioFocus LIF² system is capable of detecting picomolar concentrations of DNA, which is as sensitive as autoradiography, without hazardous radioisotopes. And fluorescent labeling of DNA fragments is easy with the dsDNA Fluorescent Detection and Analysis Kit.

Automated DNA Analysis
Simply load the carousels with buffers and samples, program the run conditions, and the BioFocus system does it all—real time detection, automatic data integration, and report printing for each analysis. Not only are capillary separations fast, but the resolution is outstanding.

Eliminate Gel Casting, Loading, and Scanning
Why spend valuable time cleaning plates, mixing buffers, casting gels, loading samples, electrophoresing, drying gels, or exposing and scanning film? BioFocus capillary separations take advantage of a ready-to-use polymer solution that creates a dynamic sieving effect. The polymer solution is automatically replenished prior to each analysis, insuring reproducibility of migration times and peak areas.

Let the BioFocus LIF^2 system work for you.

SPECIFICATIONS

Number of lasers	One or two
System configurations	1 laser/1 data channel 1 laser/2 data channels 2 lasers/2 data channels
Lasers	488 nm Argon-ion laser, 594 nm Helium Neon yellow laser
Laser operating power	Argon-ion, 3.5 mW; HeNe, 1 mW
Excitation wavelengths	488 nm, 594 nm
Detection sensitivity	10.0 x 10^{-12} M Fluorescein
Dimensions	42 cm x 31 cm x 61 cm (H x W x D)
Weight	34 kg

Ordering Information

	CATALOG NO.	PRODUCT DESCRIPTION
	LIF^2 Detector Configurations	
New	148-1200	BioFocus LIF^2 Detector, includes 488 nm argon-ion laser module and 1 data collection channel
New	148-1201	BioFocus LIF^2 Detector, includes 488 nm argon-ion laser module and 2 data collection channels
New	148-1202	BioFocus LIF^2 Detector, includes 488 nm argon-ion laser, 594 nm helium neon laser module and 2 data collection channels
	LIF^2 Detector Accessories	
New	148-1210	BioFocus LIF^2 Upgrade Kit, for BioFocus 3000 and 2000 systems
New	148-1211	BioFocus LIF^2 Upgrade Kit, for BioFocus 3000 (pre-1996 system)
New	148-3054	BioFocus LIF^2 Fluorescent User Assembled Capillary Cartridge
New	148-4133	CE dsDNA 1000 Fluorescent Detection and Analysis Kit

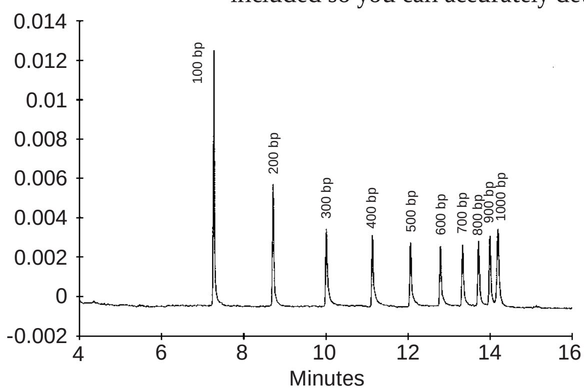
dsDNA 1000 Fluorescent Detection and Analysis Kit

Fluorescent DNA Analysis

Bio-Rad now offers a completely automated DNA analysis system that's extremely fast and convenient. Our new dsDNA 1000 Fluorescent Detection and Analysis Kit is designed to complement the new BioFocus capillary electrophoresis Laser Induced Fluorescence LIF^2 Detection System. These two comprise an excellent choice for automated fluorescent analysis of DNA fragment size polymorphisms, DNA point mutations, microsatellite DNA analysis, DNA deletion analysis, and PCR* product analysis.

Easy Labeling, High Resolution

The new dsDNA 1000 Fluorescent Detection and Analysis Kit contains all the reagents you need to quickly label and resolve double stranded DNA fragments ranging from 100 to 1,000 base pairs. The kit utilizes a special hydrophilic polymer buffer which creates a sieving effect analogous to an acrylamide gel. The ready-to-use buffer is replenished prior to each analysis, insuring the highest reproducibility of DNA fragment migration time and peak area. A 100 base-pair DNA ladder is included so you can accurately determine fragment sizes.



Bio-Rad 100 bp molecular ruler (170-8202) resolved with the ds DNA Fluorescent Detection and Analysis Kit.

SPECIFICATIONS

Fragment size range	100–1,000 bp
Resolution	10–20 bp
Number assays	400 runs
Capillary	BioCAP 75 µm ID, coated, 50 cm length, 2
Dye	Fluorescent nucleic acid stain
Run buffer	Hydrophilic polymer in TBE
DNA standard	100 bp ladder
Dye excitation wavelength	488 nm
Dye emission wavelength	520 nm

Superior Fluorescence Sensitivity.

The BioFocus LIF^2 detector is at least 1,000 times more sensitive than existing UV absorbance detection. The key to laser induced fluorescence sensitivity is focusing a beam of high intensity, single wavelength laser light on a capillary. The BioFocus LIF^2 detector is capable of detecting picomolar concentrations of DNA, which is as sensitive as autoradiography, without the hazardous radioisotopes.

Electrophoretic Analysis that's Completely Automated.

Simply load the carousels with buffers and samples, program the run conditions, and the BioFocus system does it all—real time detection, automatic data integration, and report printing for each DNA analysis. Not only are capillary separations fast, but the resolution is outstanding.

The BioFocus system virtually eliminates the need for conventional separation techniques including gel casting, sample loading, staining, exposing film, and film imaging.

BioFocus LIF^2 is the most advanced capillary electrophoresis system. Let the BioFocus LIF^2 system work for you! Contact your Bio-Rad representative today.

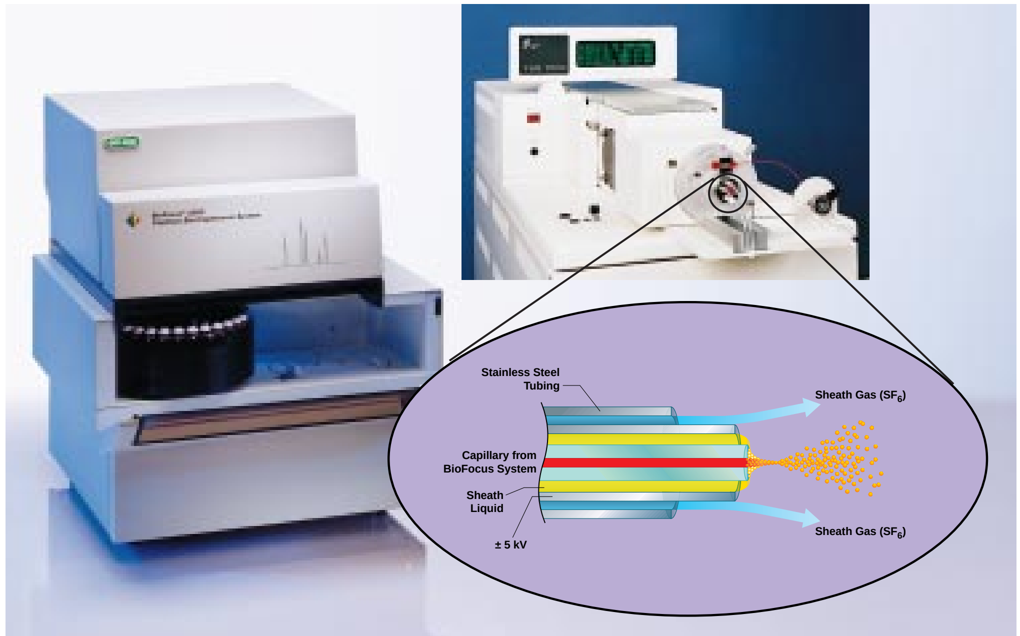
The Polymerase Chain Reaction (PCR) Process is covered by patents owned by Hoffman-LaRoche. Use of the PCR process requires a license.

Ordering Information

	CATALOG NO.	PRODUCT DESCRIPTION
New	148-4132	dsDNA 1000 Fluorescent Detection and Analysis Kit, includes 2 capillaries, run buffer, DNA ladder, fluorescent nucleic acid stain
New	148-5100	dsDNA 1000 Fluorescent Nucleic Acid Stain

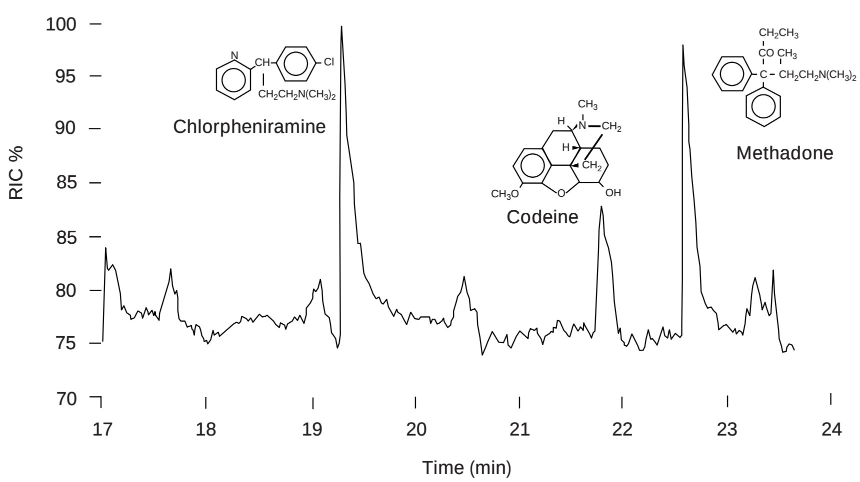
BioFocus CE/Mass Spectrometer Interface Kit

The BioFocus CE/MS interface kit connects the BioFocus CE system to a mass spectrometer with an electrospray ionization head. This interface combines the high resolving power of capillary electrophoresis with the identifying power of mass spectrometry.



Simultaneous UV/Vis and Mass Detection

The UV/Vis absorbance detectors of the BioFocus systems are fully functional while running in tandem with the mass spectrometer. High speed UV/Vis scanning on the BioFocus 3000 system produces spectral data for additional confirmation of peak identity.



Relative ion current electropherogram from an electrophoresis run with the BioFocus 3000 system operating in tandem with a Finnigan TSQ 700 mass spectrometer fitted with an electrospray interface. The BioFocus CE/MS interface kit linked the BioFocus system to the Finnigan electrospray head. Conditions: capillary, 50 µm ID x 365 µm OD BioCAP coated capillary with a total length of 75 cm; inlet buffer, 20 mM acetic acid, pH 3.2; sheath liquid, 70/29.5/0.5% isopropanol/water/formic acid delivered at a flow rate of 4.3 µl/min; injection, 5 kV, 5 sec with electrospray voltage off during injection; separation electric field, 273 V/cm; electrospray voltage, +4.5 kV.

Easy Field Upgrade

All BioFocus systems are easily upgraded to accept the MS interface in the field in a matter of hours. The interface allows quick conversion between regular capillary electrophoresis function and CE/MS. The kit is compatible with mass spectrometers fitted with the Finnigan API electrospray head.

Ordering Information

CATALOG NO.	PRODUCT DESCRIPTION
BioFocus CE/Mass Spectrometer Interface Kit	
148-1100*	BioFocus CE/MS Interface Kit with Hydraulic Cart, for connection to the Finnigan API electrospray head
148-1101*	BioFocus CE/MS Interface Kit, for connection to the Finnigan API electrospray head
Accessories	
148-1102	Hydraulic Cart

**Inquire for information on additional models of mass spectrometers compatible with this kit.*

CE Capillaries and Cartridges

The BioFocus capillary cartridge allows the insertion of any 360–375 µm OD capillary from 24–100 cm total length. BioCAP capillary columns come in a variety of sizes, with or without an internal coating.

Capillary Selection

Three variables must be considered when choosing a capillary: length, internal diameter (ID), and internal surface. Selection of the proper capillary for an application is determined by the effect that these parameters have on the performance of the system and the quality of the separation.

Increasing the length of the capillary increases the resolution of the separation, but also increases the time of the analysis, requires higher voltages, and increases the resistance to fluid flow when filling or purging the capillary. Generally, it is desirable to select the shortest capillary that will achieve the resolution needed.

Increasing the ID of the capillary increases the path length of the detector and therefore the sensitivity. It also increases the current and joule heating within the capillary, which may cause thermal distortion of the analyte bands and a decrease in resolution. For fraction collection purposes, a larger ID is desirable to increase the amount of sample injected and separated. Larger ID capillaries (50 µm or greater) are also required when using highly viscous polymer solutions for sieving applications.

The internal surface of a capillary can be modified to reduce sample adsorption to the wall and reduce or control electro-osmotic flow (EOF). Bio-Rad's coated capillaries are very effective for both of these purposes.

BioCAP Capillary Columns

BioCAP high quality capillary columns are made from capillaries drawn to precise dimensions from ultra pure fused silica and clad with a polyimide jacket for strength and flexibility. Each column comes in a case with a specially designed mounting to protect the fragile detection window.

BioCAP Coated Capillary Columns

The BioCAP coated capillary columns have an internal coating covalently bonded to the silica.* This coating offers several important advantages. Analyte adsorption to the capillary wall is greatly reduced, allowing high pH analysis of many proteins that cannot be accomplished on an uncoated capillary. Electro-osmotic flow (EOF) in the capillary is virtually eliminated, allowing the use of shorter capillaries at lower total voltages and resulting in faster, sharper, more reproducible separations.

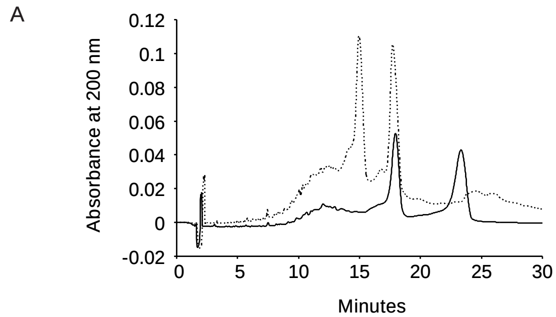
Protein adsorption is a common problem in capillary electrophoresis of proteins, particularly under alkaline conditions. Successive separations of a protein mixture were performed on an uncoated capillary at pH 8.5. Adsorption of proteins to the fused silica surface caused variable EOF, poor peak shape, and poor reproducibility of migration time and peak area. With the coating, both adsorption and EOF are greatly reduced, allowing reproducible protein separations.

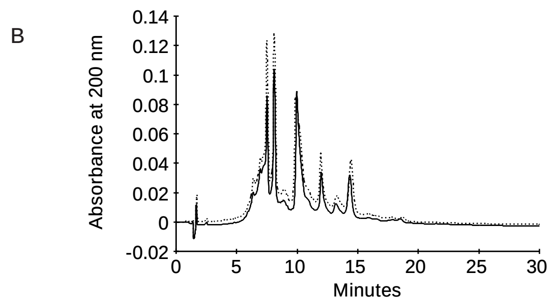
Conditions

Capillary:	*24 cm x 50 µm, A. uncoated, B. coated*
Buffer:	*Basic protein analysis*
Injection:	*8 kV for 8 seconds*
Polarity:	*Negative to positive*
Run voltage:	*8 kV*
Detection:	*200 nm*
Sample:	*β-lactoglobulin* *β-lactoglobulin B* *α-lactalbumin* *Bovine Hb* *Phycocyanin*

1st run - - - - - - - -

4th run ————





Four successive runs of a protein mixture at pH 8.5 in a A. Uncoated capillary B. Coated capillary. Only the first and fourth are shown.

Capillary Cartridge Assembly Kit

The Capillary Cartridge Assembly Kit allows you to build a BioFocus cartridge with BioCAP capillaries, or capillaries from other sources. Capillaries with dimensions of 360–375 µm OD, and between 24 and 100 cm in length, can be accommodated.

Capillary Selection Guide

CATALOG NO.	DESCRIPTION (internal diameter x length)	USES
148-3060	BioCAP bare silica capillary, 50 µm ID x 40 cm	MEKC, Low and high viscosity buffers for all uncoated applications
148-3061	BioCAP bare silica capillary, 75 µm ID x 100 cm	MEKC, Low and high viscosity buffer for all uncoated applications
148-3062	BioCAP bare silica capillary, 50 µm ID x 100 cm	MEKC, Low and high viscosity buffers for uncoated applications
148-3070	BioCAP LPA coated capillary, 50 µm ID x 100 cm	Low and high pH CZE for peptides and proteins, capillary IEF, dynamic sieving
148-3071	BioCAP LPA coated capillary, 75 µm ID x 100 cm	Dynamic sieving for nucleic acids and proteins
148-3080	Oligonucleotide analysis capillary, 75 µm ID x 30 cm, AAEE coating	Oligonucleotide separations
148-3100	BioCAP bare silica capillary, 50 µm x 10 m	MEKC, low and high viscosity buffers for all uncoated applications
148-3101	BioCAP bare silica capillary, 75 µm x 10 m	MEKC, low and high viscosity buffers for all uncoated applications
148-3102	BioCAP bare silica capillary, 100 µm x 10 m	MEKC, low and high viscosity buffers for all uncoated applications

* *US patent number 4,680,201*

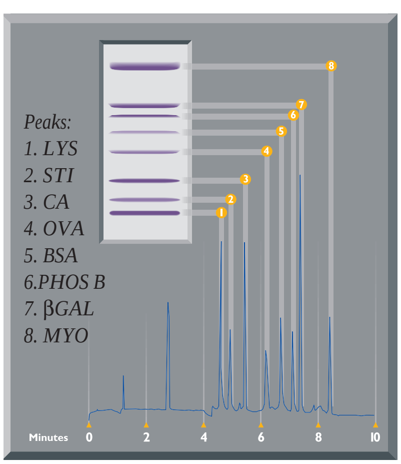
Ordering Information

CATALOG NO.	PRODUCT DESCRIPTION
BioFocus Capillary Cartridges	
148-3050	Capillary Cartridge Assembly Kit, includes spare parts
148-3051	Capillary Cartridge Spare Parts Kit
148-3052	Capillary Cartridge, 50 cm x 50 µm, uncoated, capillary installed
148-3053	Capillary Cartridge, 24 cm x 50 µm, uncoated, capillary installed
148-6072	Inlet Seal Fitting, 5
148-6073	Inlet Barbed Fitting, 5
148-6074	Outlet Barbed Fitting, 5
148-6075	Silicone Seal, red, 10
148-6076	Silicone Seal, white, 10
148-6077	Tubing, polyurethane for capillary coolant, 12 feet (4 meters)
148-6084	Alignment Tool
BioCAP Capillary Columns	
148-3060	BioCAP Bare Silica Capillary, 50 µm ID x 375 µm OD x 40 cm, window at 8 cm, 2
148-3061	BioCAP Bare Silica Capillary, 75 µm ID x 375 µm OD x 1 m, window at 60 cm
148-3062	BioCAP Bare Silica Capillary, 50 µm ID x 375 µm OD x 1 m, window at 60 cm
148-3070	BioCAP LPA Coated Capillary, 50 µm ID x 375 µm OD x 1 m, window at 60 cm
148-3071	BioCAP LPA Coated Capillary, 75 µm ID x 375 µm OD x 1 m, window at 60 cm
148-3080	BioCAP Oligonucleotide Analysis Capillary, 75 mm ID x 375 mm OD x 30 cm, window at 8 cm, 2
148-3100	BioCAP Bare Silica Capillary, 50 µm ID x 375 µm OD x 10 m, no window
148-3101	BioCAP Bare Silica Capillary, 75 µm ID x 375 µm OD x 10 m, no window
148-3102	BioCAP Bare Silica Capillary, 100 µm ID x 375 µm OD x 10 m, no window

Application Kits and Reagents

CE-SDS Protein Kit

Molecular weight determination of proteins using traditional SDS-PAGE is a time consuming, manual, multi-step process with limited quantitative accuracy. Capillary electrophoresis provides a rapid method to separate, quantify, and determine the molecular weight of proteins from 14–200 kD. Minor protein components can be detected and quantified at levels below 1% of total protein content.



The CE-SDS Protein Kit uses a polymer solution to create a dynamic sieving effect within a capillary that is analogous to the crosslinked polyacrylamide sieving used in conventional slab gel electrophoresis. With CE-SDS, the separation can be done with less sample, fewer reagents, and in a fraction of the time required for SDS-PAGE.

The CE-SDS Protein Kit contains two 50 µm ID capillaries and enough reagents to run 300 protein samples. Reagents include sample preparation buffer, dynamic sieving run buffer, protein size standards, and an internal reference standard.

The correlation between CE-SDS migration time and SDS-PAGE migration distance is illustrated (left) for the CE-SDS Protein Size Standards. Linearity of log molecular weight versus peak migration time allows generation of a calibration curve for estimating protein molecular weight.

AUTOMATED

The BioFocus capillary electrophoresis system combined with the CE-SDS Protein Kit and BioSize software provides fast, fully automated sample analysis and data handling. The dynamic sieving buffer in the capillary is replaced automatically prior to each analysis. No capillary conditioning time is required. Direct on-line UV detection provides peak migration times for molecular weight determination and peak areas for quantitation. This eliminates staining/destaining steps and the need for separate instrumentation for quantitation.

REPRODUCIBLE

Results obtained with the CE-SDS Protein Kit are not only quick and quantitative, but also highly reproducible.

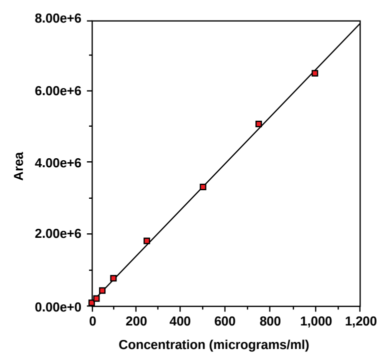
REPRODUCIBILITY OF CE-SDS

	Migration Time RSD%	Peak Area RSD%
Lysozyme	0.30	0.7
Trypsin inhibitor	0.35	1.0
Carbonic anhydrase	0.33	0.9
Ovalbumin	0.37	0.5
Serum albumin	0.39	1.0
Phosphorylase B	0.40	1.1
β-Galactosidase	0.43	1.0
Myosin	0.47	0.9

Data normalized to internal reference standard, n=10

QUANTITATIVE

On a graph showing the linearity of the integrated peak areas versus the concentration of a standard protein, carbonic anhydrase, the minimum detectable concentration is 0.36 µg/ml at a signal to noise ratio of 3.



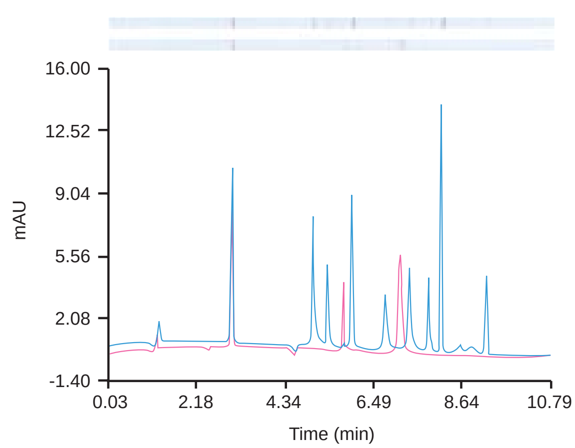
The quantitative linear range of the CE-SDS Protein Kit extends from 0.5 to 1,000 µg/ml.

APPLICATIONS

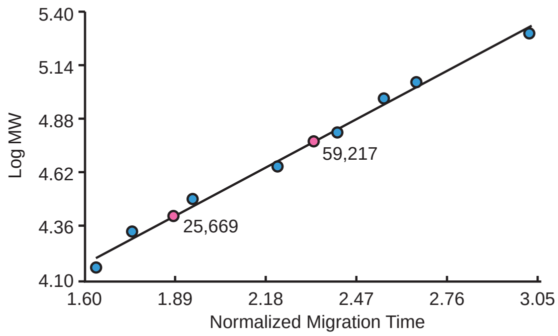
CE-SDS can be applied to many proteins and has been used for the analysis of enzymes, protein hormones, collagens, cereal proteins, immunoglobulins, protein conjugates, and recombinant protein pharmaceuticals.

BioSize Molecular Weight Estimation Software with SimGel

The BioSize software program estimates the molecular weights of sample proteins using a curve generated from the migration times of the protein standards. The program runs under Microsoft Windows and works in conjunction with the CE-SDS Protein Kit and the BioFocus integration software. This software provides a fast, convenient method for estimating the molecular weights of proteins having minimal glycosylation.



Separation of the CE-SDS Protein Standards (blue trace) and reduced murine IgG (pink trace) using the CE-SDS Protein Kit with the BioFocus automated capillary electrophoresis system. At the top of the figure, simulated gel patterns were generated from the electropherogram peaks using Bio-Rad's SimGel software package; band intensities in the simulated gel patterns are proportional to the electropherogram peak heights.



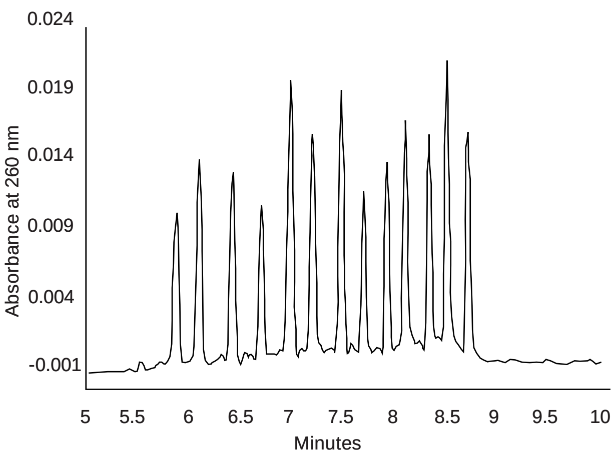
Estimation of protein molecular weight using the BioSize molecular weight determination software. The calibration plot of log MW vs migration time was generated using data from the separation of standards (blue points). Interpolated points for the monoclonal antibody peaks from the IgG separation are printed on the plot (red points) and the calculated molecular weight values are printed in red.

Using the BioSize software, the migration time data from the separation of reduced murine IgG was used to generate a graph. Molecular weight estimates for the light and heavy chains are shown to be 25,669 Da and 59,217 Da respectively.

Also illustrated is the use of the SimGel viewing utility that comes with the BioSize software (previous page). The SimGel software uses the data from the electropherogram to simulate the band patterns observed in slab gel electrophoresis. Peak heights on the electropherogram correlate directly with band intensity. These simulated bands are displayed directly above the electropherogram. This software provides users whose backgrounds are in slab gel electrophoresis with an alternative, but familiar, view of their capillary electrophoresis patterns.

CE Oligonucleotide Analysis Kit

Quality control of synthetic oligonucleotide preparations has traditionally been done by slab gel electrophoresis or HPLC, but these methods may require expensive reagents and laborious protocols, and may not provide desired quantitative precision or automation. Capillary electrophoresis is being used increasingly for determination of oligonucleotide purity because of its high resolution, low consumption of sample and reagents, and its ability to be fully automated.



Separation of the 8–32 mer mixed-sequence oligonucleotides in the CE Oligonucleotide Calibrator Set by dynamic sieving capillary electrophoresis using the CE Oligonucleotide Analysis Buffer.

The CE Oligonucleotide Analysis Kit is used for separation of oligonucleotides from 4 to 40 base units in length with baseline resolution. Separations are achieved by dynamic sieving capillary electrophoresis. In this technique, incorporation of a hydrophilic polymer in the electrophoresis buffer causes a molecular sieving effect in which nucleic acids move according to their size, with small oligonucleotides migrating faster and longer oligonucleotides migrating more slowly. The polymer type and concentration have been optimized to provide unit resolution of single-stranded oligonucleotides in the 4–40 mer range. The polymer-containing buffer is replenished between each analysis, which provides high reproducibility of migration time and peak area. The analysis employs capillaries coated internally, which enables high resolution separation of oligonucleotides of widely varying length and base composition. The kit can be applied to determination of the purity of synthetic oligonucleotides with quantitative estimation of the amounts of product and failure sequences.

The CE Oligonucleotide Kit is optimized for use with the BioFocus system. The kit can be used with other CE systems but may require modification of the analysis parameters provided in the instructions. The kit contains all components for the analysis including two lengths of BioCAP oligonucleotide analysis capillary, CE Oligonucleotide Analysis Buffer, a CE Oligonucleotide size standard containing 13 mixed-sequence oligonucleotides in lengths of 8–32 bases in 2 mer increments and instructions.

Capillary Isoelectric Focusing Reagents

Capillary isoelectric focusing (CIEF) is a high resolution technique for separation of proteins based on their isoelectric points (pI). The technique provides separations similar to those obtained with conventional gel isoelectric focusing, but is easier to perform, can be fully automated, and has the potential for providing quantitative information.

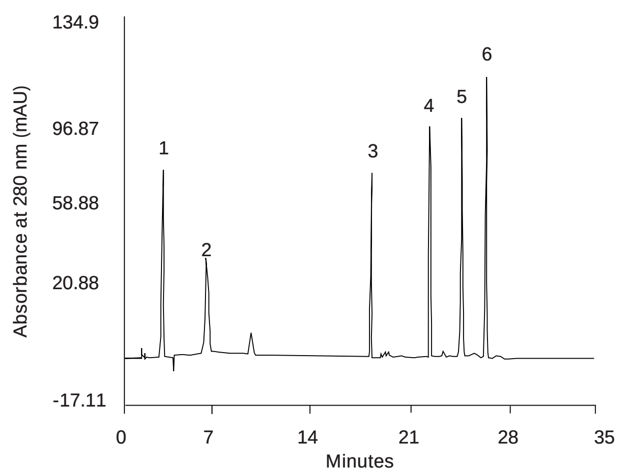
Bio-Rad offers CIEF reagents that allow high resolution separation of proteins based on their isoelectric points. In this type of analysis, a coated capillary is filled with a mixture of the sample, the CIEF Bio-Lyte ampholytes, and if desired, BioMark synthetic pI markers. A pH gradient is created by placing the acidic CIEF Anolyte in the anode reservoir and the basic CIEF Catholyte in the cathode reservoir and applying an electrical potential. After the proteins have become focused, they are mobilized past the detector by replacing the catholyte with the CIEF Mobilizer.

For accurate determination of protein isoelectric points and for achieving best reproducibility, the use of isoelectric point markers is suggested. Pure proteins of known pI values have previously been used for this purpose, but in many cases it is difficult to obtain highly pure preparations of the desired marker proteins and the proteins may degrade with storage and use, generating interfering degradation products and variable detector response.

BioMark Synthetic pI Markers for Capillary IEF

The BioMark synthetic pI markers are low molecular weight amphoteric compounds which possess well-defined isoelectric points and strong UV absorbance. They can be used as external standards for calibration of the CE system and estimation of protein isoelectric points and can be added to a protein sample as internal standards for accurate determination or confirmation of protein pI values. The markers are very stable and exhibit no degradation under conditions of normal use and storage. These highly-colored compounds are easily detected at the 280 nm wavelength typically used for CIEF of proteins, even when diluted 100-fold in the sample/ampholyte solution. In addition, their absorption in the visible region can be used to confirm peak identity when the markers are used as internal standards in complex protein mixtures.

The BioMark pI marker kit contains a 5-component mixture of markers with pI values ranging from 5.3 to 10.4, a set of 16 individual standards covering the same ranges, and instructions describing use of the markers and suggested capillary isoelectric focusing protocols for the BioFocus system.



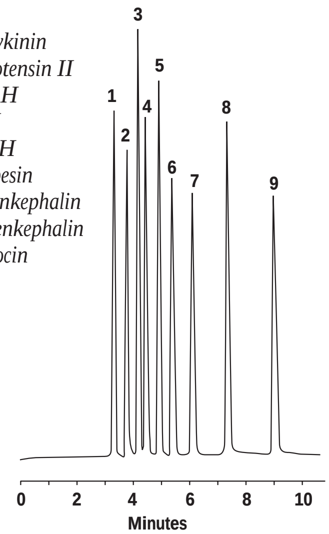
Separation of the 5-component BioMark Synthetic pI Marker mixture. The mixture was separated on the BioFocus 3000 CE system using ion addition mobilization with conditions as described in the text. Peak identities: 1. focusing peak; 2. pI 10.4 marker; 3. pI 8.4 marker; 4. pI 7.4 marker, 5. pI 6.4 marker; 6. pI 5.3 marker

Peptide Analysis Kit

Capillary zone electrophoresis provides high resolution separations of peptides. In many cases, peptides with small differences in molecular structure can be readily resolved in a matter of minutes. The best peptide separations are usually achieved under acidic conditions, typically below pH 3.

The Peptide Analysis Kit includes pH 2.5 Phosphate Buffer, a Peptide Calibration Set, and a protocol for performing peptide separations. The kit is useful for applications including purity determinations of synthetic peptides, separations of peptide digests, and characterization of peptide fractions recovered from HPLC separations. The electrolyte supplied with this kit, 0.1 M phosphate buffer, pH 2.5, is formulated for optimal performance with Bio-Rad's coated capillaries.

1. Bradykinin
2. Angiotensin II
3. a-MSH
4. TRH
5. LHRH
6. Bombesin
7. Leu-enkephalin
8. Met-enkephalin
9. Oxytocin



Electropherogram of Peptide Calibration Set.

Bio-Rad Laboratories BioFocus Integrator v 5.20

BioFocus System VERIFICATION REPORT

Report Date	Thu Jun 27 15:47:56 1996
Instrument Model	BioFocus 3000TC
Instrument Serial #	953BR011
Instrument Hardware ID	0101
Z180 Firmware Rev.	J
HC11 Firmware Rev.	F
Operating Software	Version 5.20
Operator Name	John Doe
Institution	Bio-Rad Laboratorie 2000 Alfred Nobel Drive Hercules, CA 94547

Comments

OPERATING PARAMETERS

Capillary	24 cm x 50 um uncoated
Capillary Temp	22 deg C
Sample	Verification Test Sample
Inlet	148-5010 phosphate, 0.1M
Outlet	148-5010 phosphate, 0.1M
Prep Cycle 1	Pre-Inject; Inlet, PO4_PREP; Outlet, Waste; 30 sec HI Pressure
Injection	8 psi*sec
Run Voltage	8.00 kV - to +
Detector	Single wavelength at 200 nm
Carousel Temp	23 deg C

TEST DATA

File Name	Migration Time	Peak Area	Date/Time
VER_001	3.16	702026	Wed Apr 03 15:18:34 1996
VER_002	3.18	688470	Wed Apr 03 15:26:42 1996
VER_003	3.18	691011	Wed Apr 03 15:34:50 1996
VER_004	3.18	688740	Wed Apr 03 15:42:58 1996
VER_005	3.19	684701	Wed Apr 03 15:51:06 1996
VER_006	3.19	688591	Wed Apr 03 15:59:14 1996
VER_007	3.18	688083	Wed Apr 03 16:07:22 1996
VER_008	3.18	691826	Wed Apr 03 16:15:32 1996
VER_009	3.18	685213	Wed Apr 03 16:23:38 1996
VER_010	3.19	691193	Wed Apr 03 16:31:48 1996
CALCULATED MEAN	3.180	689985	
CALCULATED STD DEV	0.008	4841	
CALCULATED RSD	0.25%	0.70%	
ACCEPTABLE RSD	1.50%	2.50%	
RESULTS	PASS	PASS	

Signature:______________________ Date:__________

The Peptide Calibration Set is used to demonstrate the peptide analysis protocol and assess the operation of the BioFocus systems. It contains a mixture of 25 µg each of the following nine peptides in lyophilized form: bradykinin, angiotensin II, melanocyte stimulating hormone, thyrotropin releasing hormone, luteinizing hormone-releasing hormone, bombesin, leucine enkephalin, methionine enkephalin, and oxytocin. After proper dilution, the final concentration of each peptide is 50 µg/ml.

New BioFocus Verification Kit

The BioFocus verification kit provides a simple way to verify the performance of the BioFocus system. Included are two capillaries, run buffer, and test sample. When used as instructed, the test compound produces a single sharp peak during each of 10 runs. Consistency of peak migration time and peak area determine system performance. BioFocus software version 5.2 or higher has a pre-installed analysis template and integration method for running and analyzing the verification test and automatically generating a verification report.

Ordering Information

CATALOG NO.	PRODUCT DESCRIPTION
Application Kits	
148-4110	Peptide Analysis Kit, includes pH 2.5 phosphate buffer, Peptide Calibration Set, and detailed analysis protocol; no capillary cartridge included
New 148-4140	CE Oligonucleotide Analysis Kit, includes 2 capillaries, run buffer, test sample, and protocol for use on BioFocus Systems
148-4160	CE-SDS Protein Kit, includes 2 uncoated capillaries (40 cm x 50 µm ID x 375 µm OD, window at 8 cm) run buffer, sample preparation buffer, internal marker, protein size standards, acidic and basic wash solutions and instructions
New 148-4170	BioFocus Verification Kit, includes 2 uncoated capillaries (40 cm x 50 µm ID x 375 µm OD, window at 8 cm) run buffer, test sample and instructions
148-4501	BioSize with SimGel, molecular weight estimation software for use with BioFocus integration software and the CE-SDS Protein Kit
Buffers	
148-5010	Phosphate Buffer, 0.1 M, pH 2.5, 60 ml x 4
148-5011	Phosphate Buffer, 0.1 M, pH 2.5, 250 ml
148-5023	Basic Protein Analysis Buffer, pH 8.5, 250 ml
148-5024	PCR Analysis Buffer, pH 8.3 Tris borate, 250 ml
148-5025	PCR Analysis Buffer with 7 M Urea, pH 8.3 Tris borate, 250 ml
148-5026	CE Oligonucleotide Buffer
148-5032	CE-SDS Protein Run Buffer
148-5033	CE-SDS Protein Sample Buffer
IEF Electrolytes	
148-5028	CIEF Catholyte, sodium hydroxide, 60 ml x 4
148-5029	CIEF Anolyte, phosphoric acid, 60 ml x 4
148-5030	CIEF Mobilizer, 60 ml x 4
148-5031	CIEF Bio-Lyte Ampholyte, pH 3–10, 2%, 60 ml
BioMark Synthetic pI Markers	
148-2100	BioMark Synthetic pI Marker Kit
148-2101	BioMark Synthetic pI Markers, pI 5.3, 6.4, 7.4, 8.4, 10.4, 200 µl
148-2102	BioMark Synthetic pI Marker, pI 5.3, 200 µl
138-2103	BioMark Synthetic pI Marker, pI 6.2, 200 µl
148-2104	BioMark Synthetic pI Marker, pI 6.4, 200 µl
148-2105	BioMark Synthetic pI Marker, pI 6.5, 200 µl
148-2106	BioMark Synthetic pI Marker, pI 6.6, 200 µl
148-2107	BioMark Synthetic pI Marker, pI 7.0, 200 µl
148-2108	BioMark Synthetic pI Marker, pI 7.2, 200 µl
148-2109	BioMark Synthetic pI Marker, pI 7.4, 200 µl
148-2110	BioMark Synthetic pI Marker, pI 7.5, 200 µl
148-2111	BioMark Synthetic pI Marker, pI 7.7, 200 µl
148-2112	BioMark Synthetic pI Marker, pI 7.9, 200 µl
148-2113	BioMark Synthetic pI Marker, pI 8.4, 200 µl
148-2114	BioMark Synthetic pI Marker, pI 8.5, 200 µl
148-2115	BioMark Synthetic pI Marker, pI 8.6, 200 µl
148-2116	BioMark Synthetic pI Marker, pI 10.1, 200 µl
148-2117	BioMark Synthetic pI Marker, pI 10.4, 200 µl
Capillary Wash Solution	
148-5022	Capillary Wash Solution, 60 ml
Standards	
148-2012	Peptide Calibration Set
148-2013	Protein Calibration Set
148-2014	CE Oligonucleotide Size Standard
148-2015	CE-SDS Protein Size Standard, 14–200 kD
148-2016	CE-SDS Internal Reference
170-3465	Low Range DNA Size Standards
161-0310	IEF Standards

CE Application Notes Index

Capillary Electrophoresis Application Notes*

MOLECULE	APPLICATION NOTE	DESCRIPTION
Glutathione	1575-1	CZE separation of oxidized and reduced forms of glutathione
Hemoglobin	1575-2	IEF separation of hemoglobin variants A, F, S, and C
PCR product	1575-4	Dynamic sieving separation of 500 base pair PCR product from reaction components
Pentapeptides	1575-7	Resolution of reversed peptide sequences by CZE
Substance P	1575-8	CZE separation of substance P peptide sequences
Tear proteins	1575-10	Dynamic sieving separation of proteins found in human tears
Caffeine	1575-11	Analysis of caffeine in beverages
hGH	1575-12	Resolution of intact and deamidated human growth hormone
Nucleotides	1575-14	CZE separation of nucleoside triphosphates
Vitamin C	1575-15	CZE separation of ascorbic acid from isoascorbic acid
Protein standards	1575-16	Protein standards separated under alkaline conditions
BSA	1575-18	Comparison of human serum albumin separated by IEF and CZE
HSA	1575-19	Tryptic peptides separated on coated and uncoated capillaries
B vitamins	1575-22	MEKC separation of water-soluble vitamins
Food dyes	1575-23	MEKC separation of food dyes
Casein & Whey	1575-25	CZE separation of milk proteins
Porcine Insulin	1575-28	Effect of pH on CZE separation of porcine insulin
Rare earth	1575-29	Separation of rare earth metals with indirect detection
IEF of protein standards	1575-31	Capillary IEF of protein standards
Dipeptides	1575-38	CZE separation of closely related dipeptides
Sandostatin	1575-39	CZE separation of Sandostatin sequences
Human insulin	1575-41	CZE analysis of recombinant human insulin
TRH	1575-42	CZE separation of TRH analogs and degradation products
Organic acids	1575-43	Analysis of organic acids in wine with indirect detection
Daunorubicin	1575-44	MEKC separation of daunorubicin with scanning detection
Antibiotics	1575-45	CZE separation of sulfonamide antibiotics
Amphotericin B	1575-46	MEKC separation of amphotericin B with scanning detection
Spider venoms	1575-47	CZE analysis of spider venoms
Organic acids	1575-49	Analysis of hop bitter acids by MEKC with scanning detection
Enantiomers	1575-51	Separation of Sympathomimetic Drug Enantiomers by CE
Nucleic acids	1575-52	Analysis of Amplified Nucleic Acids by CE using Polymer Solutions
Lead in food	1575-53	Analysis of Lead in Foods using CE
Peptides	1575-54	Analysis of Peptides by CE/MS
Drugs	1575-55	Analysis of Chlorpheniramine, Codeine, and Methadone by CE/MS

**New Application Notes are added to our library regularly. Contact your Bio-Rad Representative for updates.*

Group Headquarters • Bio-Rad Laboratories • 2000 Alfred Nobel Drive •
Hercules California 94547
(510) 741-1000 FAX (510) 741-5800
Toll Free: 1-(800) 4-BIORAD 1 (800) 879-2289
[1-(800) 424-6723] Telex: 335-358

Australia • Bio-Rad Laboratories Pty., Ltd. • Block Y, Unit 1 • Regents Park Industrial Estate • 391 Park Road • Regents Park, New South Wales 2143
02 9914 2800 FAX 02 9914 2889
Toll Free: 1800-224 354 (within Australia only)

Austria • Bio-Rad Laboratories Ges.m.b.H. • Auhofstraße 78D • A-1130 Wien
(1) 877 89 01 FAX (1) 876 56 29

Belgium • Bio-Rad Laboratories S.A.-N.V. • Begoniastraat 5 • B-9810 Nazareth EKE
09-385 55 11 FAX 09-385 65 54

Canada • Bio-Rad Laboratories (Canada) Ltd. • 5671 McAdam Road •
Mississauga Ontario L4Z 1N9
(905) 712-2771 FAX (905) 712-2990
Toll Free: 1 (800) 268-0213

Peoples Republic of China • Bio-Rad China (Beijing) •
14 Zhi Chun Road, Hai Dian District • Beijing 100 008
86-10-2046622 FAX 86-10-2051876

France • Bio-Rad S.A. • 94/96 rue Victor Hugo • B.P. 220 • 94 203 Ivry Sur Seine Cedex
(01) 43 90 46 90 FAX (01) 46 71 24 67

Germany • Bio-Rad Laboratories GmbH • Heidemannstraße 164 • D-80939 München •
Postfach 45 01 33 • D-80901 München
089 318 84-0 FAX 089 318 84-100

Hong Kong • Bio-Rad Pacific Ltd. • Unit 1111, 11/F., New Kowloon Plaza •
38 Tai Kok Tsui Road • Tai Kok Tsui, Kowloon
852-2789-3300 FAX 852-2789-1257

India • Bio-Rad Laboratories (India) Pvt. Ltd. • C-248 Defence Colony • New Delhi 110 024
(91-11) 461 0103 FAX (91-11) 461 0765

Israel • 12 Homa Street • P.O. Box 5076 • Rishon Le Zion 75655
03-951 4127 FAX 03-951 4129

Italy • Bio-Rad Laboratories S.r.l. • Via Cellini, 18/A • 20090 Segrate - Milano
02 21609.1 FAX 02 21609.399

Japan • Nippon Bio-Rad Laboratories KK • 7-18 Higashi-Nippori 5-chome, •
Arakawa-ku Tokyo 116
03-5811-6270 FAX 03-5811-6272

Korea • Hangwoo Buildig 6F, 1342-3 • Seocho-Dong • Seocho-Gu • Seoul
03-5811-6270 FAX 03-5811-6272

Netherlands • Bio-Rad Laboratories B.V. • Fokkerstraat 10 • 3905 KV Veenendaal
31 318-540666 FAX 31 318-542216

New Zealand • Bio-Rad Laboratories Pty Ltd. • Unit 15 Poland Court • 21 Poland Road •
P.O. Box 100-051 • North Shore Mail Centre • Glenfield Auckland 10
09-443 3099 FAX 09-443 3097
Toll Free: 0508-805 500 (within New Zealand only)

Singapore • Bio-Rad Laboratories, Pte Ltd. (Singapore) • 221 Henderson Road •
05-19 Henderson Building • Singapore 0315
65-2729877 FAX 65-2734835

Spain • Bio-Rad Laboratories S.A. • Avda Dr Severo Ochoa, Edificio B2–Piso 3 •
Parque Casablanca • Poligono Industrial de Alcobendas • 28100 Alcobendas (Madrid)
(91) 661 70 85 FAX (91) 661 96 98
FreePhone: 900 100 204

Sweden • Bio-Rad Laboratories AB • Ekensbergsvägen 128 • Box 1097 • S-172 22 Sundbyberg
46 (0)8 627 50 00 FAX 46 (0)8 627 54 00
Freephone: 46 020-660 660

Switzerland • Bio-Rad Laboratories AG • Kanalstrasse 17 • Postfach • CH-8152 Glattbrugg
01-809 55 55 FAX 01-809 55 00

United Kingdom • Bio-Rad Laboratories Ltd. • Bio-Rad House • Maylands Avenue •
Hemel Hempstead • Hertfordshire HP2 7TD
01442-232552 FAX 01442-259118
FreePhone:0800-181134

BIO-RAD

Printed in USA Bulletin 1955 REV B